

CUSTOM R–XL FAULT IMPEDANCE BANK

Design Basis, Calculation & Implementation Note

System Voltage: 3 V RMS, 50 Hz	Nominal Current: Up to 1 kA	Status: Proposed Design
Target Range: $R = 0\text{--}3\text{ m}\Omega$, $L = 0\text{--}9\text{ }\mu\text{H}$	Control: 4-Bit Binary (16 Steps)	Scale Target: 10 kA-Class

1. PURPOSE

The proposed high-current short-circuit test system requires a programmable low-impedance network capable of controlling:

- Fault current magnitude (I)
- Circuit resistance (R)
- Circuit inductive reactance (X_L)
- Resulting power factor ($\text{PF} = \cos \phi$)

For the scaled prototype, a custom binary-weighted R–XL bank is proposed instead of commercially available resistor/inductor modules. The bank is designed around the prototype operating point of **3 V RMS, 50 Hz, up to 1 kA**. The design serves as a scaled proof-of-concept for the proposed 10 kA-class architecture.

2. DESIGN REQUIREMENTS

For a target fault current I , the required loop impedance magnitude is:

$$|Z| = \frac{V}{I} \quad (1)$$

At the maximum prototype test current ($V = 3\text{ V}$, $I = 1000\text{ A}$):

$$|Z| = \frac{3}{1000} = 3\text{ m}\Omega \quad (2)$$

Therefore, the complete fault network must provide an impedance adjustable across the milliohm range. For a specified power factor ($\text{PF} = \cos \phi$):

$$R = |Z| \cdot \text{PF}, \quad X_L = |Z| \sqrt{1 - \text{PF}^2}, \quad L = \frac{X_L}{2\pi f} \quad (3)$$

where:

- R = circuit resistance (Ω)
- X_L = inductive reactance (Ω)
- L = circuit inductance (H)
- $f = 50\text{ Hz}$ = test system frequency

3. CUSTOM R BANK

A binary-weighted resistance network is proposed using discrete series elements:

Bank Element	Target Resistance	Function
R_1	0.2 m Ω	Bit 1 (LSB)
R_2	0.4 m Ω	Bit 2
R_3	0.8 m Ω	Bit 3
R_4	1.6 m Ω	Bit 4 (MSB)
Maximum	3.0 mΩ	Full series sum (16 selectable states)

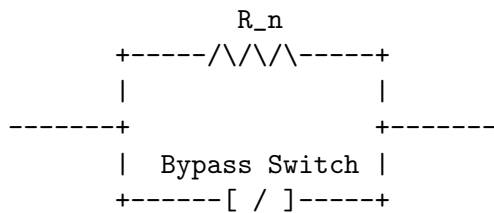
The total bank resistance is defined as:

$$R_{\text{bank}} = b_1(0.2) + b_2(0.4) + b_3(0.8) + b_4(1.6) \quad [\text{m}\Omega] \quad (4)$$

where each $b_n \in \{0, 1\}$, providing $2^4 = 16$ possible resistance configurations.

Switching Concept

The resistance elements are connected in series. Each element can be bypassed using an individual switching device:



The bypass switches are configured prior to the fault shot and do **not** interrupt the 1 kA fault current. Fault initiation is executed by a dedicated SCR semiconductor switch.

4. R-BANK POWER REQUIREMENTS & THERMAL DESIGN

During a short-circuit fault pulse, the instantaneous resistive power dissipation is:

$$P_R = I^2 R \quad (5)$$

At $I = 1000$ A:

Resistance Element	Instantaneous Power (at 1 kA)
0.2 mΩ	200 W
0.4 mΩ	400 W
0.8 mΩ	800 W
1.6 mΩ	1.6 kW
3.0 mΩ	3.0 kW

These values represent transient fault-pulse power, not continuous ratings. The physical thermal sizing is governed by total pulse energy:

$$E_R = \int I^2 R dt \quad (6)$$

This allows the bank to be constructed from compact, pulse-rated, low-inductance metal strips or alloy meanders rather than continuous-duty commercial resistors.

5. CUSTOM X_L BANK & INDUCTANCE REQUIREMENT

The inductive section utilizes a binary-weighted air-core reactor arrangement to eliminate ferromagnetic saturation during high-current fault transients:

Bank Element	Target Inductance	Function
L_1	0.6 μH	Bit 1 (LSB)
L_2	1.2 μH	Bit 2
L_3	2.4 μH	Bit 3
L_4	4.8 μH	Bit 4 (MSB)
Maximum	9.0 μH	Full series sum (16 selectable states)

The total bank inductance is:

$$L_{\text{bank}} = b_1(0.6) + b_2(1.2) + b_3(2.4) + b_4(4.8) \quad [\mu\text{H}] \quad (7)$$

At $f = 50$ Hz and maximum inductance ($L = 9 \mu\text{H}$):

$$X_L = 2\pi f L = 2\pi(50)(9 \times 10^{-6}) \approx 2.83 \text{ m}\Omega \quad (8)$$

Because X_L is of the same order of magnitude as R_{bank} (0–3 mΩ), this provides smooth power factor control.

6. EXAMPLE OPERATING POINTS ($|Z| = 3 \text{ M}\Omega$ TARGET)

- **Target PF ≈ 0.6 :**

$$\begin{aligned} R &= 3 \times 0.6 = 1.8 \text{ m}\Omega \\ X_L &= 3 \times \sqrt{1 - 0.6^2} = 2.4 \text{ m}\Omega \\ L &= \frac{2.4 \times 10^{-3}}{2\pi(50)} \approx 7.64 \text{ }\mu\text{H} \end{aligned}$$

Binary approximation: $R = 1.8 \text{ m}\Omega$ ($R_2 + R_3 + R_4$ adjusted or $R_3 + R_4 = 2.4 \text{ m}\Omega$ / $R_1 + R_4 = 1.8 \text{ m}\Omega$), $L = 7.8 \text{ }\mu\text{H}$ ($L_2 + L_3 + L_4$).

- **Target PF ≈ 0.5 :**

$$\begin{aligned} R &= 3 \times 0.5 = 1.5 \text{ m}\Omega \\ X_L &= 3 \times \sqrt{1 - 0.5^2} \approx 2.60 \text{ m}\Omega \\ L &= \frac{2.60 \times 10^{-3}}{2\pi(50)} \approx 8.27 \text{ }\mu\text{H} \end{aligned}$$

- **Target PF ≈ 0.4 :**

$$\begin{aligned} R &= 3 \times 0.4 = 1.2 \text{ m}\Omega \\ X_L &= 3 \times \sqrt{1 - 0.4^2} \approx 2.75 \text{ m}\Omega \\ L &= \frac{2.75 \times 10^{-3}}{2\pi(50)} \approx 8.75 \text{ }\mu\text{H} \end{aligned}$$

The 0–9 μH inductive bank cleanly covers the required test envelope.

7. HIGH-CURRENT CONSTRUCTION CONSIDERATIONS

At kiloampere pulse currents, parasitic loop parameters significantly alter the test circuit. Physical layout must follow strict low-inductance, low-resistance practices:

- Flat copper busbars with minimized spacing to lower parasitic lead inductance.
- Air-core copper reactors arranged orthogonally or shielded to prevent inter-coil mutual coupling.
- Meander/serpentine resistor strips with counter-parallel current paths to minimize self-inductance.
- High-pressure bolted copper lug connections to keep contact resistance well below $0.05\text{ m}\Omega$.
- Dedicated galvanic isolation between gate-drive electronics and high-current power stages.

All nominal design values represent theoretical targets; baseline parasitics of busbars, SCR packages, and transformer secondaries must be measured and compensated.

8. CONTROL ARCHITECTURE

The system enforces a strict two-stage switching philosophy:

1. **Configuration Stage:** The operator selects target I and PF via HMI. The PLC energizes pre-selection bypass contactors under zero-current (dry switching) conditions.
2. **Fault Initiation Stage:** Zero-crossing detector synchronizes timing; the ESP32 micro-controller triggers the SCR gate via an optoisolated driver at the chosen phase angle, initiating the high-current pulse.

9. VERIFICATION PLAN & SCALE-UP

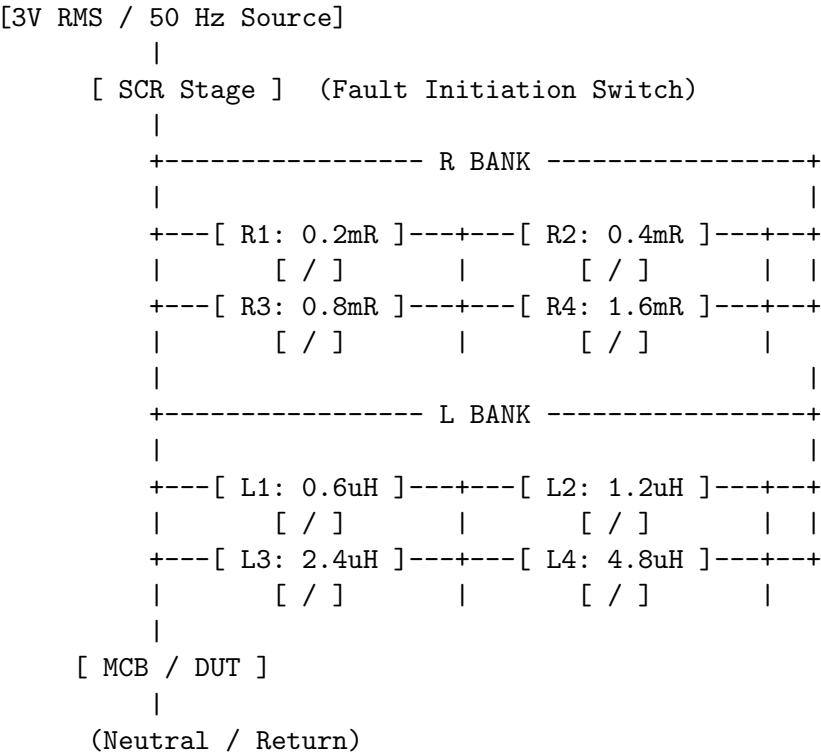
Before high-current commissioning, each module must undergo independent low-level parameter extraction:

- **R-Elements:** 4-wire Kelvin DC resistance, pulse thermal rise, contact resistance.
- **L-Elements:** Precision LCR bridge characterization at 50 Hz and harmonic frequencies.
- **Full Network:** Complete complex impedance measurement $Z = R + jX_L$ compared against calibrated baseline targets.

The architecture scales directly to 10 kA by increasing conductor cross-sections, SCR pulse ratings, and mechanical clamping stiffness without changing the binary-weighted topology.

APPENDIX A: IMPLEMENTATION SCHEMATICS & CALCULATION MATRIX

A.1 Complete System Topology



A.2 Sample Binary Impedance Matrix (50 Hz)

State	b ₄ b ₃ b ₂ b ₁ (R)	R _{bank}	b ₄ b ₃ b ₂ b ₁ (L)	L _{bank}	X _L (50 Hz)	Z Target
0x0	0000	0.0 mΩ	0000	0.0 μH	0.00 mΩ	0.00 mΩ
0x1	0001	0.2 mΩ	0011	1.8 μH	0.57 mΩ	0.60 mΩ
0x3	0011	0.6 mΩ	0101	3.6 μH	1.13 mΩ	1.28 mΩ
0x7	0111	1.4 mΩ	1000	4.8 μH	1.51 mΩ	2.06 mΩ
0x8	1000	1.6 mΩ	1100	7.2 μH	2.26 mΩ	2.77 mΩ
0xF	1111	3.0 mΩ	1111	9.0 μH	2.83 mΩ	4.12 mΩ

Note: All calculated values represent nominal design parameters. Verified values must be calibrated experimentally on the fabricated prototype before running test qualification shots.